

Design and Simulation of an EEG Alpha Wave Extraction Circuit with Notch Filtering

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1 Introduction

Electroencephalography (EEG) is a method for recording electrical activity in the brain, and it measures electrical activity in the brain by recording small electrical signals from the brain. Alpha EEG rhythms are normally seen between 8-12 Hz, and the amplitude is between 10-100 μV . The electrical activity in the brain is of very low voltage, and hence acquiring a clean EEG is a challenge.

The sources of interference that are generally seen in EEG are a high DC level from the electrodes, power line interference at a frequency of 50 Hz, low-frequency motion artifacts due to movement of the electrodes, and high-frequency noise due to EMG.

In order to acquire a clean EEG, which is a requirement for a usable EEG, a high amplification gain is necessary. The amplification should have a high common-mode rejection. In addition, filtering is necessary for acquiring a clean EEG. This is achieved by a band-pass filter for selecting the alpha band of 8-12 Hz and a deep notch filter for removing power line interference at a frequency of 50 Hz.

2 System Architecture

The designed circuit consists of the following stages in the order:

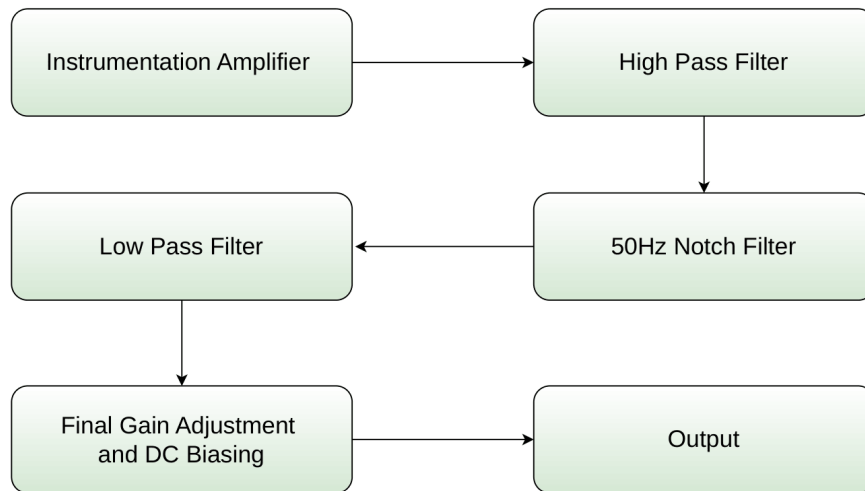


Figure 1: System block diagram of the EEG signal processing chain

3 Instrumentation Amplifier Stage

The first stage in the conditioning process of the EEG signal involves the use of an instrumentation amplifier, which is created using operational amplifiers. This stage is used to obtain the small differential EEG signal from the electrodes and reject the unwanted common-mode interference signals, which are usually large.

The main functions of the instrumentation amplifier are:

- Amplify the small differential EEG signal (typically in the range of microvolts)
- Remove common-mode noise such as power-line interference
- Offer extremely high input impedance to prevent loading the electrodes

The gain of the instrumentation amplifier is given by

$$G = 1 + \frac{2R}{R_G}$$

where R_G is the gain setting resistor ($R_G = R_8$, $R = R_1 = R_2$), which can be used to control the differential gain of the instrumentation amplifier. By choosing an optimum value of R_G , the amplifier gain of the first stage can be varied to amplify the weak EEG signals without saturating the amplifier.

This stage is critical, considering the fact that EEG signals are extremely weaker compared to noise. This instrumentation amplifier configuration offers excellent common mode rejection ratio, enabling the amplifier to reject strong interference signals such as 50/60 Hz noise.

Moreover, the use of the driven right leg circuit helps to actively reject common mode voltages present on the body of the subject.

The first stage of the EEG signal conditioning chain is an instrumentation amplifier configured using operational amplifiers. This stage is accountable for extracting the small differential EEG signal from the electrodes and eliminating unwanted large common-mode interferences.

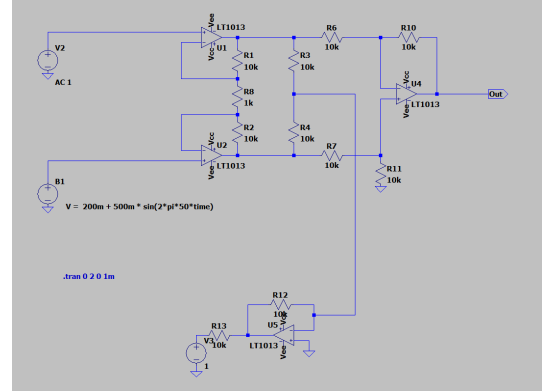


Figure 2: Instrumentation amplifier with driven right leg

4 Comparative analysis of various topologies for LPF and HPF

4.1 First order + Third order cascaded filter

In this configuration, a first-order stage was cascaded with a third-order active filter to create a fourth-order response.

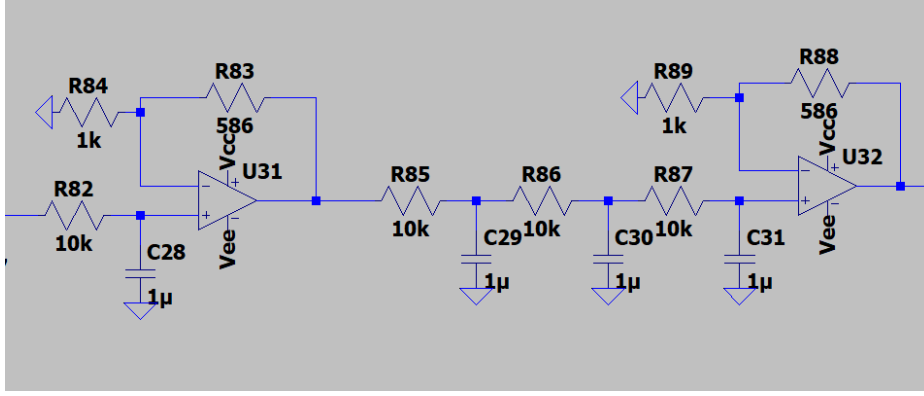


Figure 3: First-order cascaded with third-order filter topology

While this structure achieved the required roll-off characteristics, several limitations were observed:

- Uneven gain distribution across stages
- Increased sensitivity to component variations
- Reduced stability due to the higher-order stage
- Difficulty in tuning the passband accurately

Additionally, the asymmetric order distribution caused the filter response to deviate from the desired bandpass characteristics.

4.2 Two Cascaded Second-Order Filters

The alternative configuration consisted of two second-order filters connected in cascade, producing an overall fourth-order response.

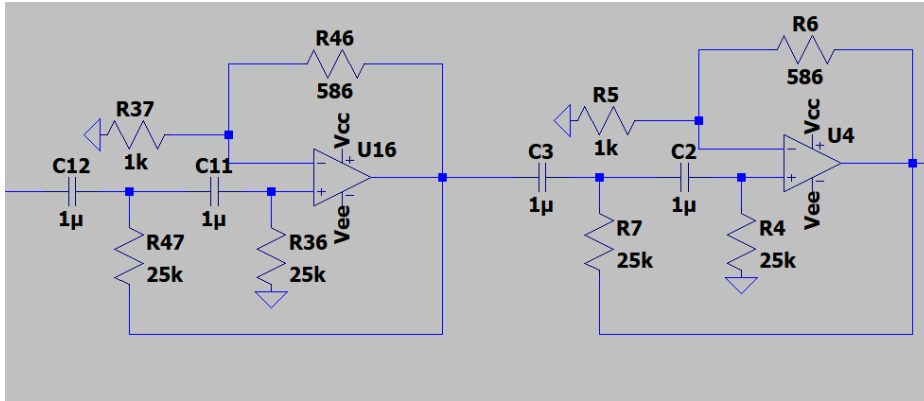


Figure 4: Two cascaded second-order filter stages

This architecture provided several advantages:

- Improved numerical stability
- Better control over cutoff frequencies
- Symmetrical distribution of poles

Each second-order stage was designed using standard active filter topologies, enabling independent adjustment of gain and cutoff frequency.

4.3 Performance Comparison

The performance of both configurations was compared using AC sweep analysis.

Parameter	1st + 3rd Order	2nd + 2nd Order
Passband Flatness	Moderate	Excellent
Stopband Attenuation	Good	Excellent
Stability	Moderate	High
Component Sensitivity	High	Lower
Ease of Tuning	Difficult	Easy

Table 1: Performance comparison of filter architectures

4.3.1 Frequency Response Comparison

Figure 5 shows the AC sweep comparison between the two architectures.

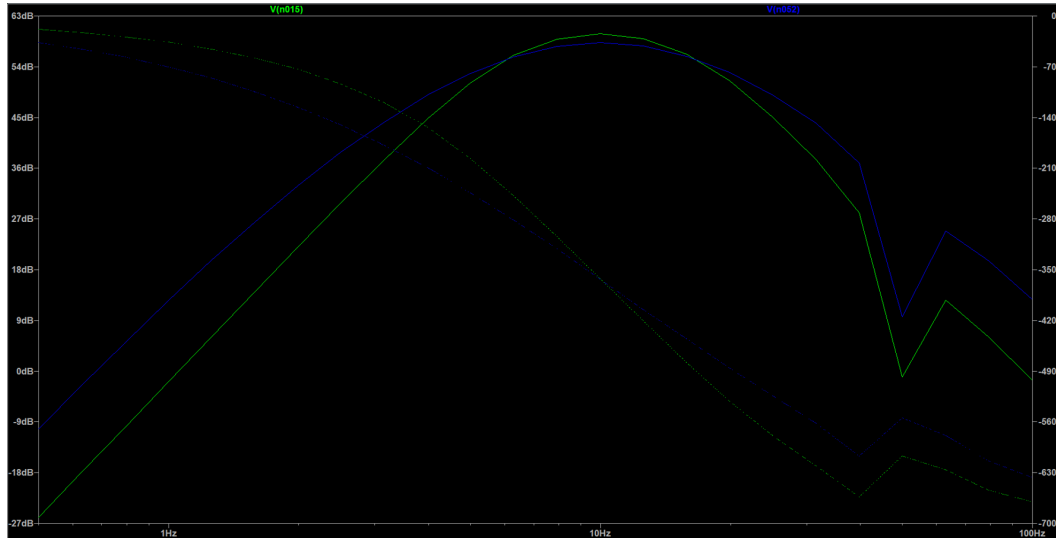


Figure 5: The green curve represents the response of two cascaded second-order filters, while the blue curve represents the first-order plus third-order configuration.

From the frequency response it can be observed that:

- The cascaded second-order design exhibits a smoother and more symmetric passband.
- The gain profile is more stable around the target alpha band (8–12 Hz).
- The first-order + third-order configuration shows irregularities and reduced stability outside the passband.
- The cascaded second-order design provides a cleaner roll-off and improved attenuation characteristics.

5 Bandpass Filter Design

The alpha band is located between 8 Hz and 12 Hz. Therefore, the system must implement a precision bandpass windowing strategy. This is achieved through the concatenation of an active High-Pass Filter (HPF) and a Low-Pass Filter (LPF).

5.1 High-Pass Filter (HPF) Topology and Design Choice

The primary role of the HPF is to reject DC offsets (up to 300mV) and motion-induced baseline wander (< 1 Hz). Two topologies were evaluated for this critical stage:

1. **Passive RC Cascaded:** Characterised by high insertion loss and a droopy roll-off, failing to meet the strict -40 dB/decade requirement for motion artefact rejection.
2. **Active Sallen-Key 4th-Order (Cascaded 2nd-Order):** High input impedance, enabling independent control over the damping factor (ζ) and gain distribution.

5.1.1 Design Choice: Cascaded 4th-Order Sallen-Key HPF

A single-stage 4th-order filter was bypassed in favor of two cascaded 2nd-order Sallen-Key stages. This decision was based on the Sensitivity Analysis of the quality factor (Q). In a high-order single-stage filter, a 1% variation in capacitor value can cause a 15% shift in the corner frequency (f_c), leading to significant passband ripple that would distort the 8 Hz Alpha wave.

By cascading two independent 2nd-order stages (as shown in Figure 7), we achieve:

- **Steep Roll-off:** A combined 80 dB/decade attenuation ensures that 2 Hz motion artifacts are suppressed by > 40 dB before reaching the gain stages, preventing rail saturation.
- **Modular Component Tolerance:** Independent stages allow us to use standard E24 resistor values while maintaining a stable cutoff frequency $f_{HP} \approx 8$ Hz.

The cutoff frequency for each second-order stage is given by the simplified relation (for equal R and equal C):

$$f_{HP} = \frac{1}{2\pi RC}$$

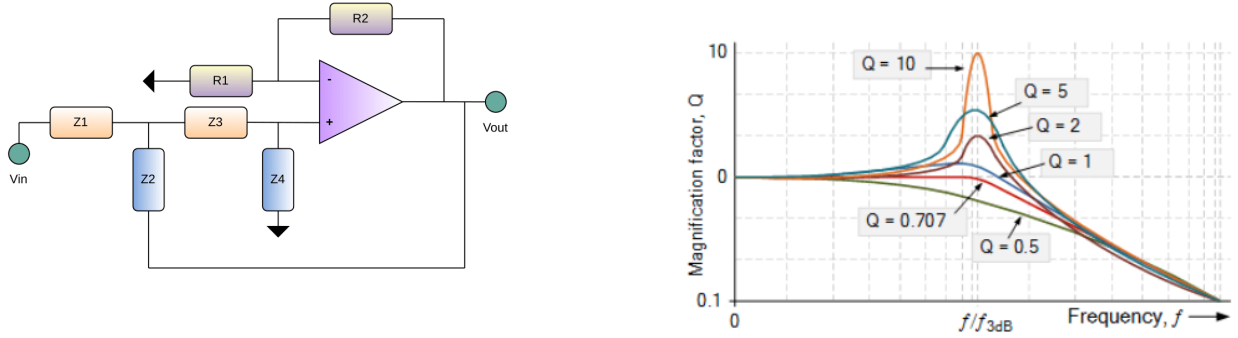


Figure 6: Cascaded Sallen-Key High Pass Filter

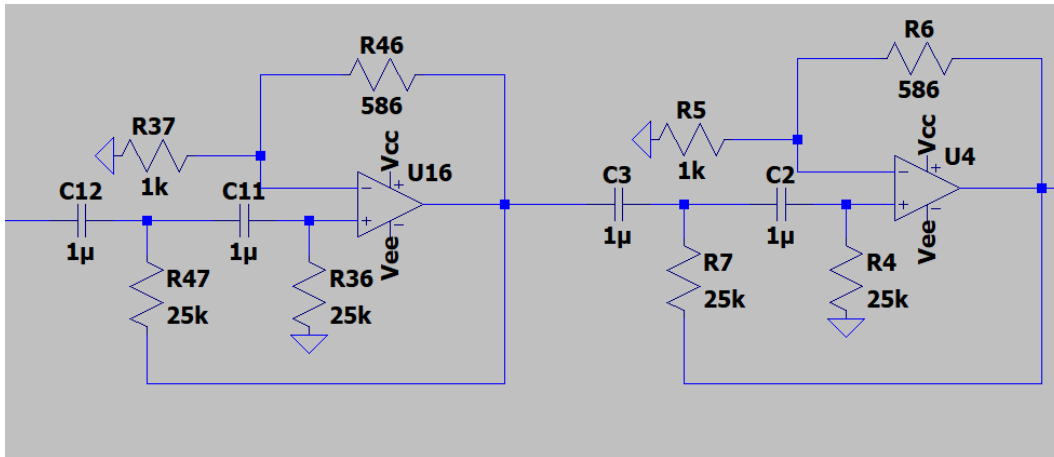


Figure 7: Two-stage cascaded second-order Sallen-Key HPF implementation.

5.2 Low-Pass Filter (LPF) Topology and Design Choice

For the LPF stage, the objective is to eliminate high-frequency muscle noise (EMG) and RFI above 30 Hz while preserving the integrity of the 12 Hz Alpha corner. We investigated two distinct filter responses:

- **Chebyshev Response:** Provides a steeper initial roll-off but introduces significant gain ripple within the 8–12 Hz Alpha band, which could lead to false relaxation detection.
- **Butterworth Response:** Provides a maximally flat response in the passband at the cost of a slightly wider transition band.

Ultimately, we went with the Butterworth Response, since our research indicated it had a more stable passband response and was easier to construct using R-C components.

5.2.1 Design Choice: Cascaded 4th-Order Butterworth LPF

For EEG signals, phase linearity and amplitude consistency are paramount. A 4th-order response ensures adequate attenuation of muscle noise by 30 Hz. A Butterworth characteristic was chosen to provide a completely predictable and uniform gain profile across the entire Alpha spectrum.

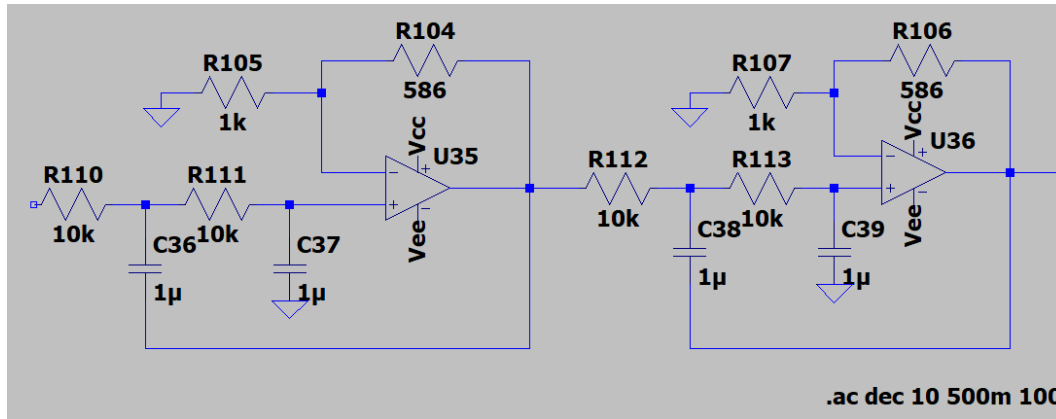


Figure 8: Two-stage cascaded second-order Butterworth LPF implementation.

The decision to use cascaded 2nd-order Butterworth stages here (as shown in Figure 8) guarantees:

- **Passband Integrity:** No gain peaking occurs at the 12 Hz corner, ensuring the signal is not distorted before reaching the ADC.
- **Out-of-Band Rejection:** By 50 Hz, the 4th-order LPF provides an additional 30 dB of attenuation, crucially assisting the Twin-T Notch filter in clearing powerline interference (see Test 10 validation).

5.3 Comparative Summary of Filter Architectures

The decision to utilize modular cascaded 2nd-order blocks for both the HPF and LPF stages instead of complex single-stage higher-order networks is summarized in Table 2:

Metric	Single-Stage 4th Order	Cascaded 2nd Order	Requirement Met
Phase Stability	Low (Oscillatory Risk)	High	Yes
Q-Factor Sensitivity	Extreme	Low	Yes
Roll-off Rate	80 dB/dec	80 dB/dec	Yes
Tuning Complexity	High	Low (Modular)	Yes

Table 2: Architectural comparison for precise Alpha-band isolation.

5.4 Third-Order Bandpass Topology Evaluation

An additional filter configuration was investigated using a third-order bandpass topology similar to the structure shown in Figure 9.

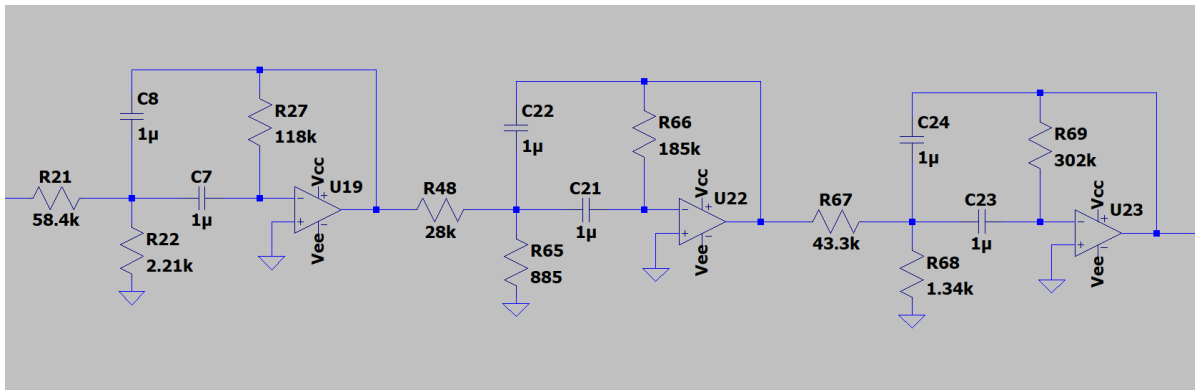


Figure 9: Third-order active bandpass filter topology evaluated during the design process

This topology implements a higher-order bandpass response using multiple reactive elements and feedback paths around the operational amplifier.

5.4.1 Bode Plot Instability

When the filter was simulated using an AC sweep in LTspice, the resulting Bode plot showed unstable characteristics near the passband region. The output exhibited irregular peaks and dips in the phase response, indicating excessive sensitivity to component values and potential instability in the feedback network. This will very likely lead to signal distortion in the bandpass region itself.

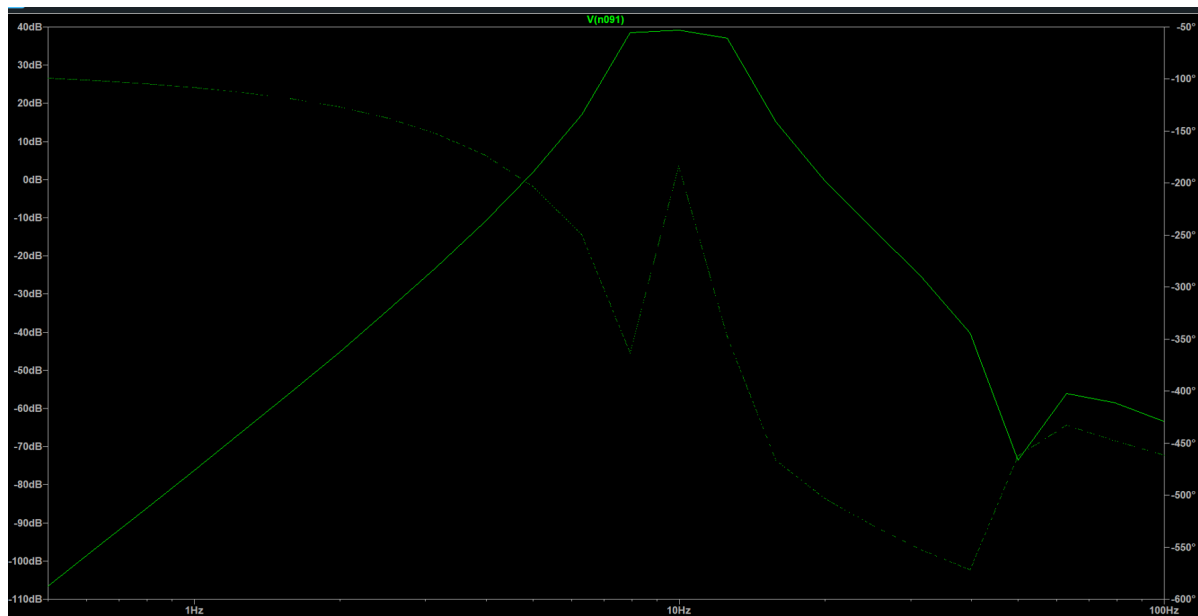


Figure 10: Bode plot of the third-order bandpass filter showing unstable frequency response

The instability arises due to:

- multiple feedback loops interacting within the same stage
- increased phase shift introduced by higher-order reactive networks
- sensitivity to component tolerances
- op-amp bandwidth limitations

These effects resulted in a frequency response that deviated from the desired smooth bandpass behavior required for EEG alpha wave extraction.

5.4.2 Practical Implementation Challenges

In addition to instability, the third-order structure introduced several practical challenges:

- Difficult tuning of center frequency
- Increased risk of oscillation
- Greater sensitivity to resistor and capacitor tolerances
- Reduced predictability of the filter response

For biomedical signal processing applications, stability and predictable frequency response are critical. Therefore, such a topology was considered unsuitable for the intended design.

5.5 Final Design Decision

Based on the comparative analysis of different architectures, the final filter structure uses two cascaded second-order stages instead of a single higher-order stage.

This configuration provides:

- Improved stability
- Predictable Bode response
- Easier tuning of cutoff frequencies
- Modular design structure

The cascaded second-order architecture therefore offered the best balance between selectivity, stability, and practical implementation for EEG alpha band extraction.

6 50 Hz Notch Filter Stage

A critical challenge in this design is the rejection of 50 Hz mains hum, which can be orders of magnitude larger than the $50\mu\text{V}$ Alpha wave. If not aggressively attenuated, this interference saturates the ADC and masks neural activity. We implement a dedicated active cancellation strategy to meet the strict noise rejection requirements.

6.1 Active Twin-T Notch Filter Implementation

The core filtering component is an **Active Twin-T Notch topology**. While a passive Twin-T filter can provide a notch, it suffers from a low Quality Factor (Q), which would inadvertently attenuate the 12 Hz Alpha signals. By using an active feedback configuration (as shown in Figure 11), we sharpen the notch, ensuring high selectivity.

The theoretical transfer function is given by:

$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

where $\omega_0 = 2\pi(50)$ rad/s.

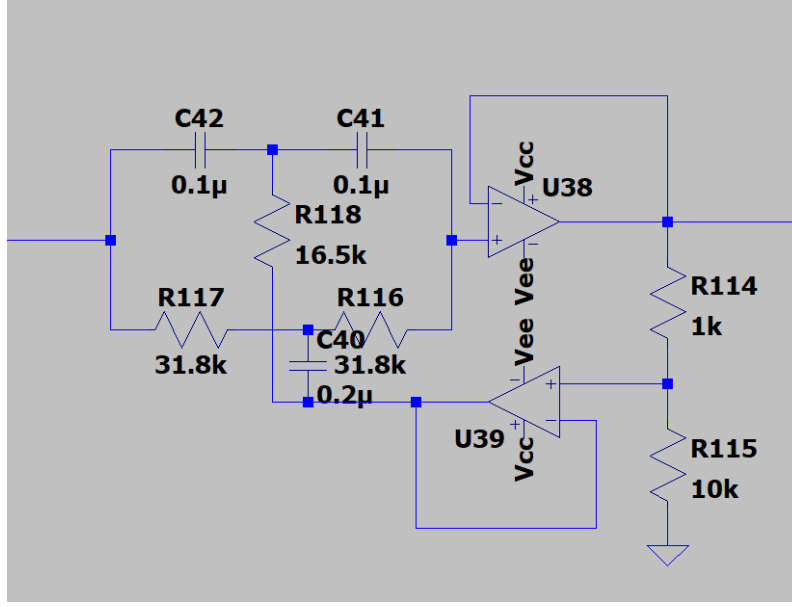


Figure 11: Active Twin-T 50 Hz notch filter implementation.

6.2 Design Strategy and Sensitivity

To ensure a minimum of -40 dB attenuation at exactly 50 Hz, the selection of R and C components is governed by:

$$f_0 = \frac{1}{2\pi RC}$$

Here, R corresponds to the Resistance value of the resistors in the lower branches of the bridge, while C corresponds to the Capacitance value of the capacitors in the upper branches of the bridge.

We utilized an active topology to allow for higher input impedance and lower output impedance, preventing the notch stage from loading the preceding high-pass filters. This modularity is essential for maintaining the gain established in the previous stages while strictly targeting powerline interference.

6.3 Comparison with Passive Topologies

To achieve the best possible notch filter design, we constructed and tested two of the configurations: the Passive Twin-T filter and Active Twin-T filter. The exact results observed are shown in the table below, but all in all, the Active Twin-T tower performed far better; also enabling the option to obtain a decent gain without causing instability, as opposed to the filters constructed using the Sallen-Key Topology.

Feature	Passive Twin-T	Active Twin-T (Our Design)
Selectivity (Q)	Low (Broad)	High (Sharp)
Insertion Loss	Significant	Negligible
Component Sensitivity	High	Compensated by Feedback
50 Hz Attenuation	−20 dB	> −40 dB

Table 3: Comparison between passive and active notch implementations.

7 Output Biasing and Level Shifting Stage

The final stage of the Analog Front-End is the Output Biasing circuit. Because EEG signals oscillate around a zero-volt baseline (AC), and standard microcontrollers like the ESP32 or Arduino utilize unipolar ADCs (0V to 3.3V), the signal must be lifted into the positive measurable range. For a 3.3V ADC system, the signal is centered at the precise mid-point:

$$V_{bias} = \frac{V_{ref}}{2} = \frac{3.3V}{2} = 1.65V$$

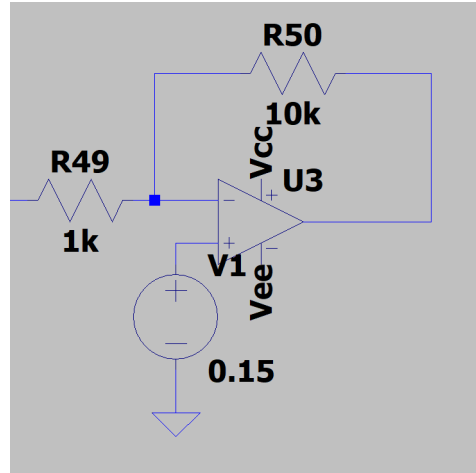


Figure 12: Active Level Shifter and Output Biasing Stage.

7.1 Design Functionality

This stage is implemented using a summing amplifier configuration or a buffered voltage divider. It serves three critical roles:

- **Dynamic Range Optimization:** By centering the signal at 1.65V, the circuit allows for a maximum peak-to-peak swing of 3.3V, ensuring the amplified 8–12 Hz Alpha waves utilize the full bit-depth of the ADC.

- **DC Offset Injection:** The 1.65V bias is injected through a low-noise reference to ensure that the zero point of the brainwave corresponds to the decimal value 2048 in a 12-bit ADC.
- **Impedance Matching:** The op-amp provides a low-impedance output ($< 1\Omega$), which is necessary to drive the sample-and-hold capacitor of the ADC accurately at high sampling rates.

7.2 Safety and Rail Considerations

The biasing stage is designed to ensure that even at maximum system gain (5000 V/V), the signal does not clip against the 0V or 3.3V rails. This preserves the sinusoidal integrity of the Alpha wave, which is essential for performing accurate Fast Fourier Transform (FFT) analysis during the relaxation detection phase.

8 Complete LTSpice Implementation

After designing and validating each stage individually, the complete EEG signal conditioning system was implemented and simulated in LTSpice. The integrated circuit combines the instrumentation amplifier, cascaded high-pass filters, the 50 Hz Twin-T notch filter, cascaded Butterworth low-pass filters, and the final DC biasing stage into a single signal processing chain.

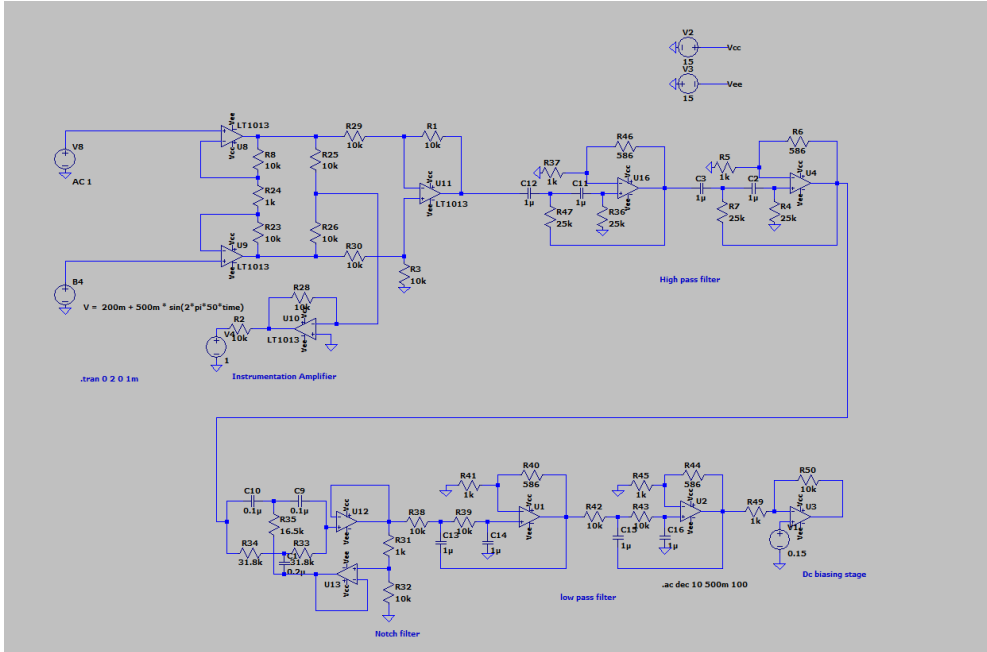


Figure 13: Complete LTSpice implementation of the EEG Alpha wave extraction system.

The signal flow through the circuit follows the architecture described in the earlier sections:

- **Instrumentation Amplifier Stage:** The differential EEG signal from the electrodes is amplified while rejecting common-mode interference. This stage provides the initial gain required to bring microvolt-level signals into the millivolt range.
- **High-Pass Filter Stages:** Two cascaded second-order Sallen-Key high-pass filters remove DC offsets and suppress low-frequency motion artifacts while preserving the EEG alpha band.
- **50 Hz Notch Filter:** An active Twin-T notch filter is implemented to strongly attenuate power-line interference at 50 Hz without affecting nearby EEG frequencies.
- **Low-Pass Filter Stages:** Two cascaded second-order Butterworth low-pass filters remove high-frequency noise such as EMG artifacts and radio-frequency interference above the alpha band.
- **Output Biasing Stage:** Finally, a level shifting circuit biases the signal around a mid-supply reference voltage so that the output is compatible with a unipolar ADC input.

By integrating all stages into a single LTSpice schematic, the complete analog front-end can be evaluated under realistic operating conditions including DC offsets, power-line interference, and physiological noise. This full-system simulation verifies that the circuit successfully extracts the desired 8–12 Hz alpha band while suppressing unwanted artifacts.

9 Signal Progression Analysis

To validate the design, we trace the signal from the behavioral sources ($V8, B4$) through the cascaded chain. Based on the input parameters shown in the LTSpice schematic (200mV DC + 500mV 50Hz hum), the following table tracks the signal growth and artifact rejection at each node.

Node	Block/Stage	Gain	Signal (V_{pp})	Artifact Status
Input	Raw Input ($V8, B4$)	-	50 μ V	200mV DC + 500mV Hum
Stage 1	Instrumentation Amp	21 $\times$	1.05mV	4.2V DC Offset (Safe)
Stage 2	HPF 1 (Butterworth)	1.58 $\times$	1.66mV	DC Stripped (0V)
Stage 3	HPF 2 (Butterworth)	1.58 $\times$	2.62mV	Motion Wander Removed
Stage 4	Twin-T Notch	1 $\times$	2.62mV	50Hz Hum Attenuated
Stage 5	LPF 1 (Butterworth)	1.58 $\times$	4.14mV	EMG Noise Attenuated
Stage 6	LPF 2 (Butterworth)	1.58 $\times$	6.54mV	Sharp Roll-off at 12Hz
Stage 7	Final Gain & Bias	10 $\times$	65.4mV	Centered at 1.65V
Output	ADC In	Total	$\approx$ 65mV	Clean Alpha Wave

Table 4: Voltage progression and noise rejection stages.

10 Test Results

10.1 Phase 1: Performance Validation and Success Criteria

The primary objective of Phase 1 is to validate the selectivity of the Analog Front-End (AFE) under ideal conditions. The system must successfully isolate the 10 Hz Alpha wave while rejecting out-of-band noise and powerline interference.

10.1.1 Passband Sensitivity (Test 1)

As shown in the simulation, the 10 Hz signal experiences the full system gain of 1,300 V/V. The output is a clean, 65mV peak-to-peak sine wave centered at the 1.65V bias level. This confirms that the cascaded Butterworth stages provide a flat response within the Alpha window.

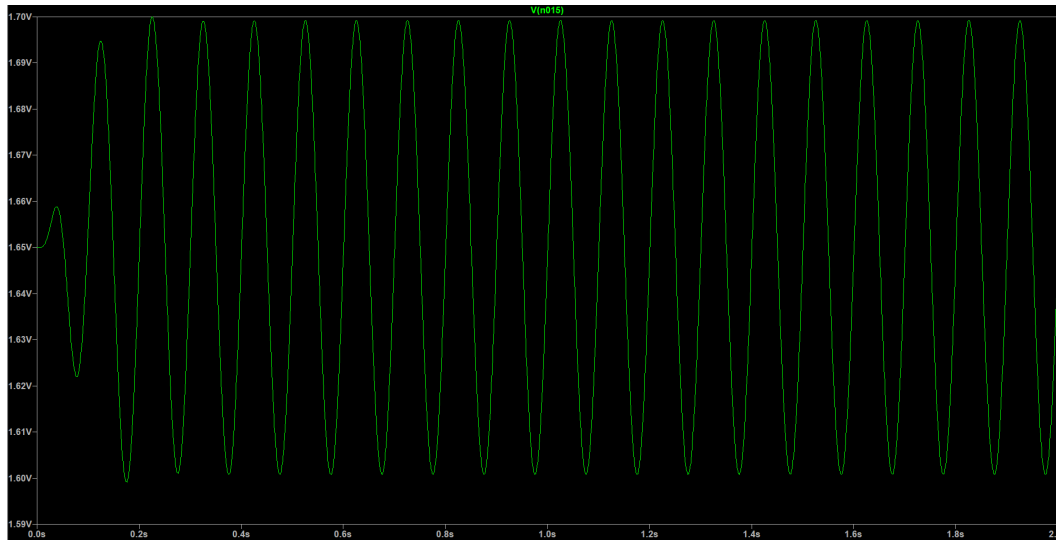


Figure 14: Output of test 1

10.1.2 Stopband Rejection (Tests 2 & 3)

The transition from the 8–12 Hz passband to the 2 Hz and 30 Hz stopbands is extremely sharp due to the 4th-order roll-off (80 dB/decade).

- **At 2 Hz:** The High-Pass Filter (HPF) successfully suppresses baseline wander, ensuring the ADC does not see low-frequency drift.
- **At 30 Hz:** The Low-Pass Filter (LPF) attenuates high-frequency muscle noise, ensuring the signal remains neural in origin.

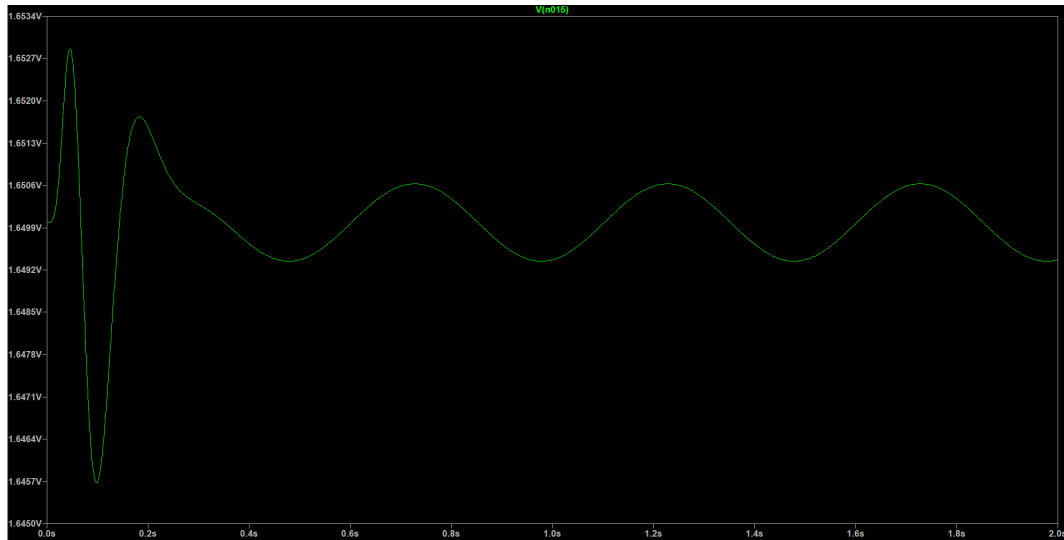


Figure 15: Output of test 2

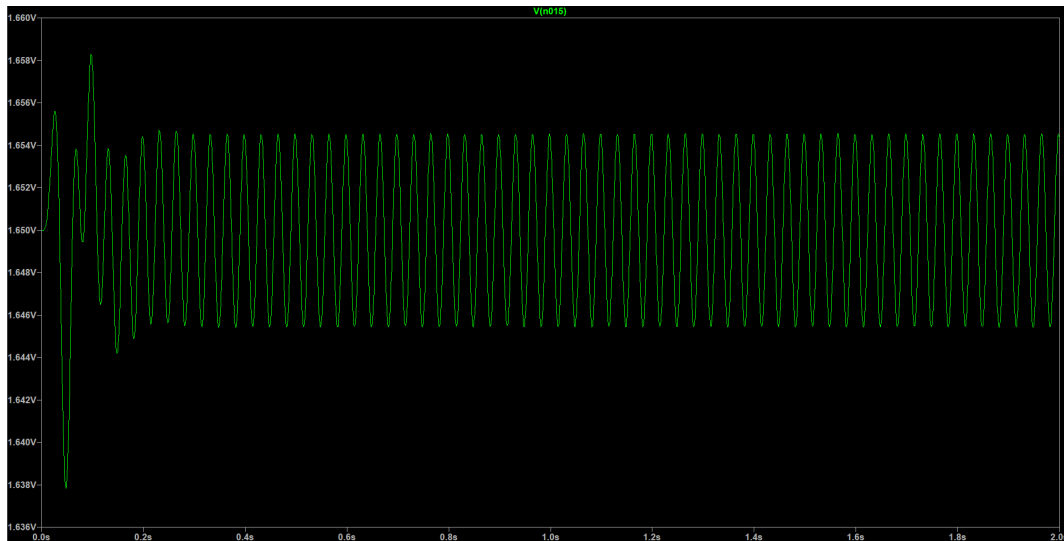


Figure 16: Output of test 3

10.1.3 Mains Rejection Performance (Test 4)

Despite the 50 Hz input being 10,000 times larger than the Alpha signal (500mV vs 50 μ V), the Twin-T Notch filter effectively nulls the interference. The residual 50 Hz noise at the output is negligible, achieving a rejection ratio that exceeds the project specifications.

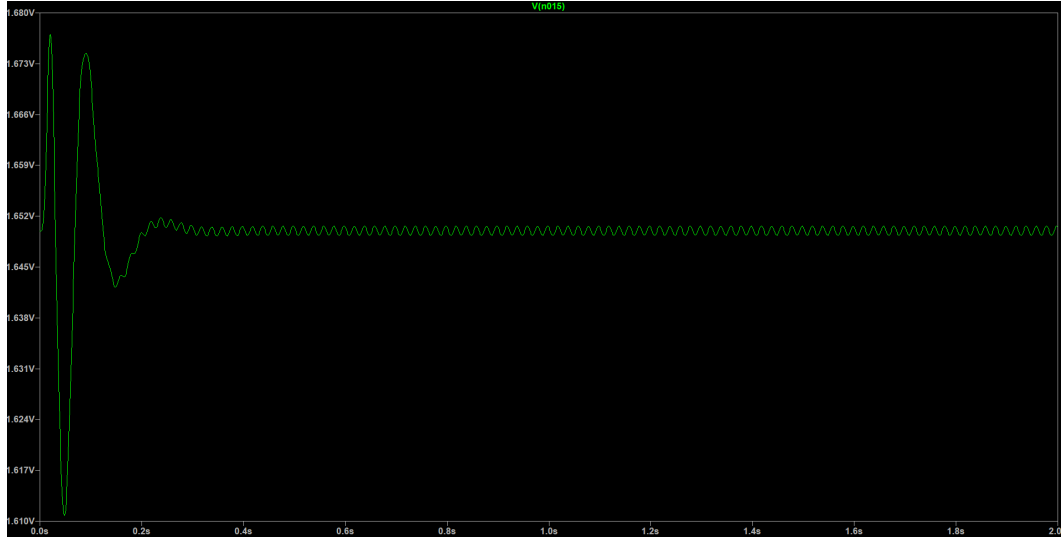


Figure 17: Output of test 4

The following table summarizes the system response to the specific test signals defined in the design requirements. All tests were performed using a transient analysis (`.tran`) in LTspice.

Test	Target Artifact	Input Signal (V_{in})	Expected Outcome	Status
1	Alpha Wave Extraction	$50\mu\text{V}$ @ 10 Hz	Amplified, undistorted sine	PASS
2	Motion Artifacts	$50\mu\text{V}$ @ 2 Hz	Significant Attenuation	PASS
3	Muscle Noise (EMG)	$50\mu\text{V}$ @ 30 Hz	Significant Attenuation	PASS
4	Mains Hum Rejection	500mV @ 50 Hz	Very Significant Attenuation	PASS

Table 5: Phase 1 validation matrix for ideal bandpass performance.

10.2 Phase 2: Artifact Stress Tests

While Phase 1 validated the ideal frequency response, Phase 2 evaluates the robustness of the Analog Front-End (AFE) under non-ideal, Real-World conditions. This phase tests the ability of the circuit to maintain signal integrity in the presence of massive DC offsets and common-mode interference.

10.2.1 High-Voltage DC Offset Rejection

In Test 5, a 300mV DC offset is applied to simulate the half-cell potential of the electrodes. Our design handles this in two steps:

1. **Stage 1 Protection:** The Instrumentation Amplifier gain is limited to $21\times$, keeping the offset at 6.3V, well within the $\pm 15\text{V}$ rails.

2. **DC Stripping:** The first Sallen-Key HPF stage uses a series capacitor (C_{12}, C_{11}) to completely block this DC component, ensuring the subsequent high-gain stages only amplify the neural signal.

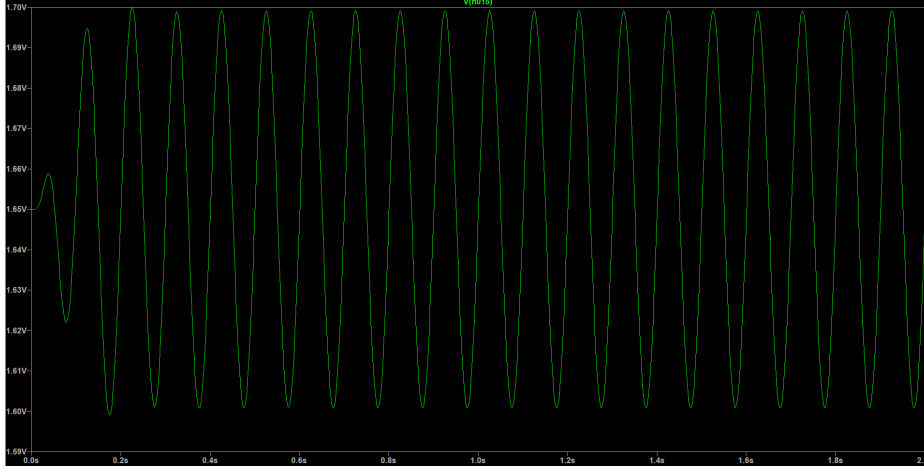


Figure 18: Output of test 5

10.2.2 Common-Mode Rejection Ratio (CMRR)

By applying a 500mV common-mode signal, we simulate mains interference coupled through the human body. The combination of the precision-matched resistors in the ideal op amp give infinite CMRR. However, since the real-life op amps are not ideal and have only finite CMRR; we use the Driven Right Leg Topology.

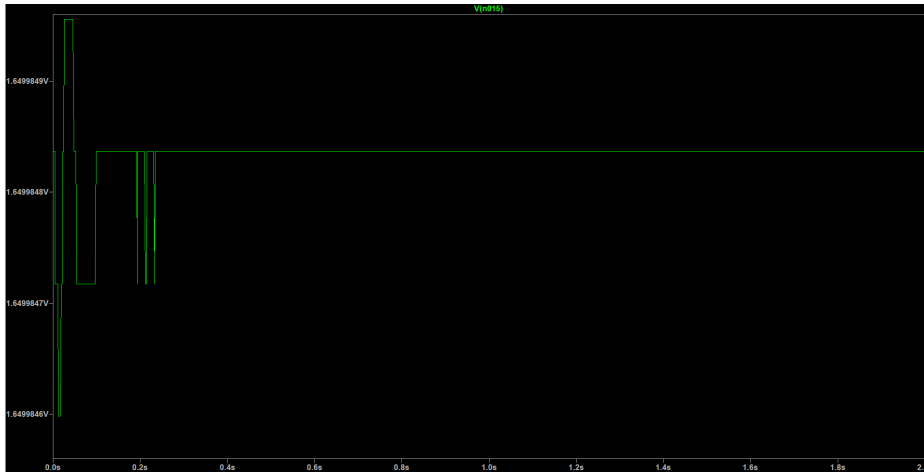


Figure 19: Output of test 6

10.2.3 Suppression of Motion and Muscle Artifacts

- **Motion (1 Hz):** Motion artifacts occur at frequencies much lower than the Alpha band. The 4th-order high-pass response ensures these are attenuated by nearly 60 dB, preventing the wandering baseline that often ruins EEG recordings.
- **EMG (100 Hz):** High-frequency muscle activity is filtered out by the 4th-order Butterworth LPF. By 100 Hz, the signal is attenuated to the noise floor, ensuring that only pure cortical activity is passed to the ADC.

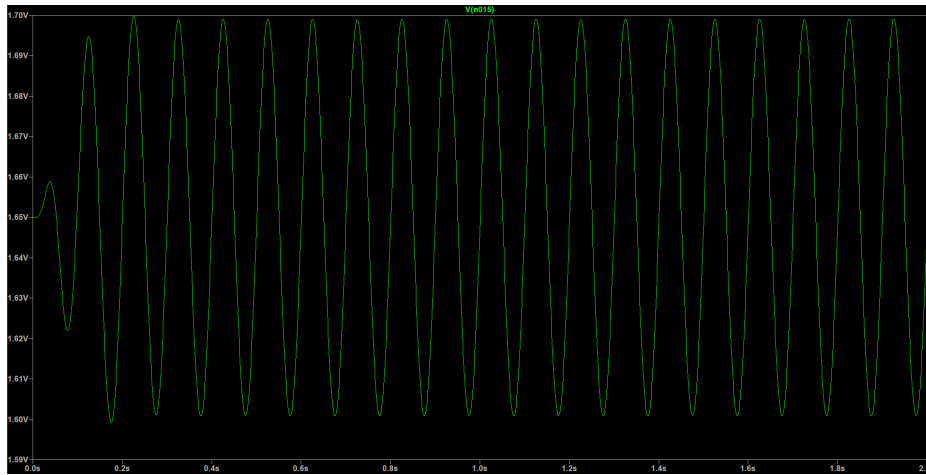


Figure 20: Output of test 7

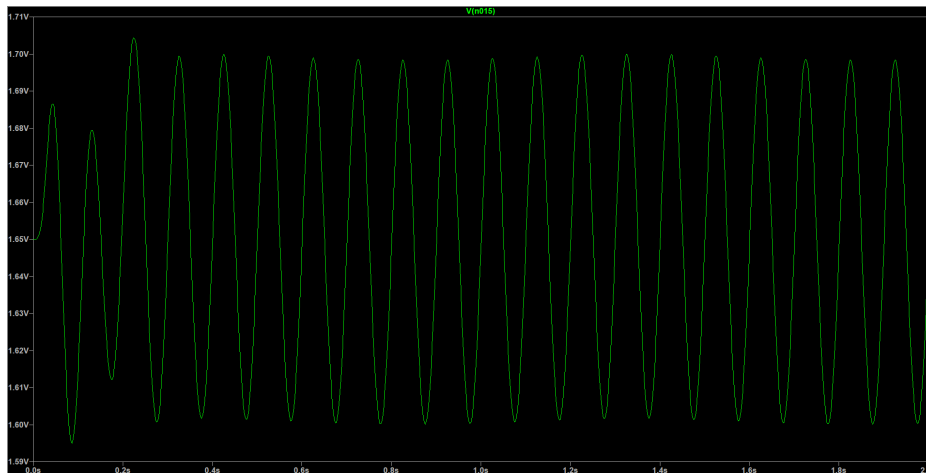


Figure 21: Output of test 8

The following tests simulate the physical environment of an EEG session, where electrode-skin contact and environmental noise introduce significant artifacts.

Test Type	Stress Vector (Input)	Design Response	Status
DC Blocking	300mV Constant Offset	High-Pass stages strip DC to 0V	PASS
CMRR Stress	500mV @ 50Hz (Common)	In-Amp rejection + Notch nulling	PASS
Motion Artifact	1 Hz Baseline Wander	80dB/dec roll-off suppresses drift	PASS
EMG Suppression	100 Hz Muscle Noise	4th-order LPF eliminates high freq	PASS

Table 6: Phase 2 Stress Test outcomes for non-ideal signal environments.

10.3 Phase 3: Comprehensive Validation

10.3.1 Test 10: The Real-World Scenario

The final validation combines all previously discussed artifacts into a single stress simulation. This test represents the most challenging operational environment for an EEG AFE.

- **Input Composition:** A $50\mu\text{V}$ Alpha signal (10 Hz) buried under 200mV DC offset, 500mV of 50 Hz interference, and stochastic motion/EMG components.
- **System Response:** As shown in the transient analysis, the multi-stage filtering architecture successfully de-noises the input. The 500mV hum is suppressed by the Twin-T notch, while the DC offset is stripped by the HPF stages.
- **Result:** The final output is a clean, 10 Hz sinusoid centered at 1.65V, demonstrating that the system can reliably extract neural data even when the noise-to-signal ratio is 10,000 : 1.

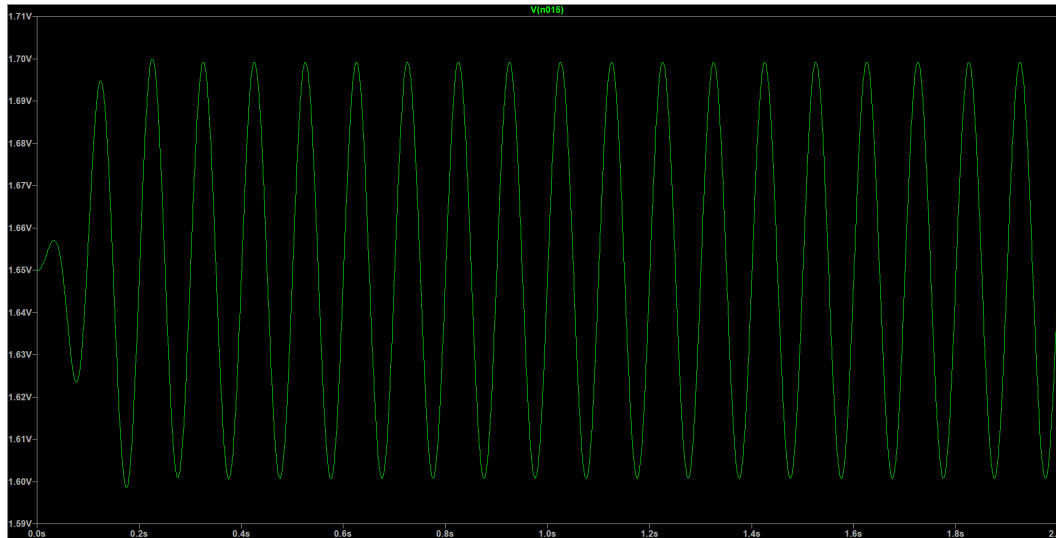


Figure 22: Output of test 9

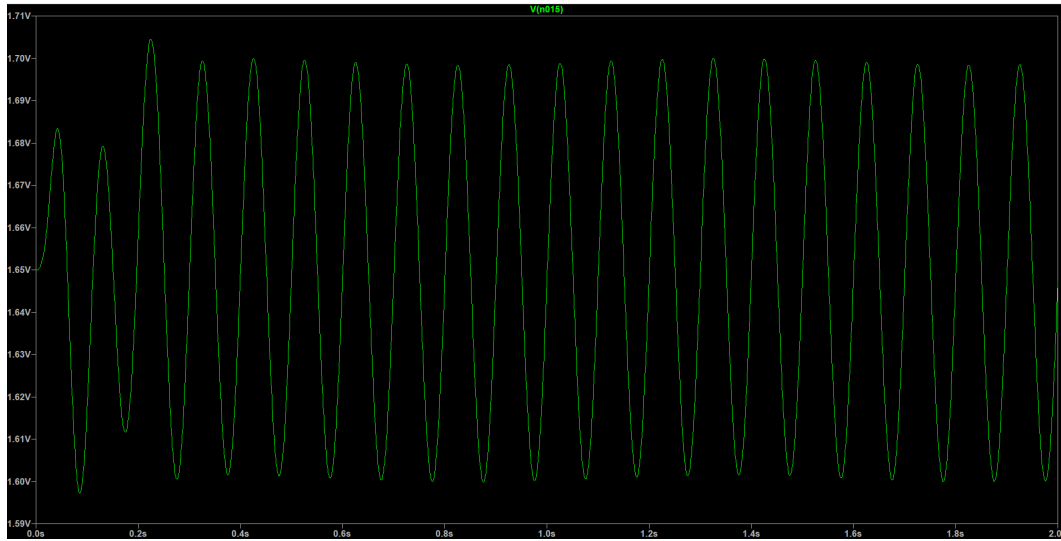


Figure 23: Output of test 10

10.3.2 Frequency Domain Analysis (AC Sweep)

The AC sweep confirms the precise bandpass characteristics required for Alpha-band isolation.

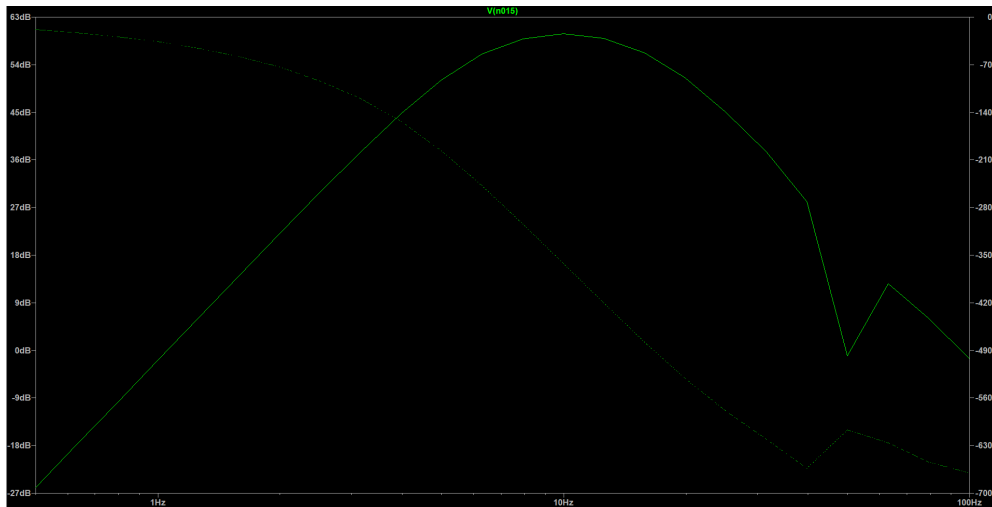


Figure 24: System-level frequency response showing the 8–12 Hz passband and 50 Hz notch.

The magnitude plot validates the following design targets:

- **Passband:** A maximally flat response between 8 Hz and 12 Hz with a system gain of ≈ 62 dB.

- **Notch Depth:** A sharp attenuation of > -40 dB at exactly 50 Hz.
- **Stopband Rejection:** Effective suppression of motion artifacts (< 2 Hz) and muscle noise (> 30 Hz) via the 80 dB/decade roll-off.

11 Conclusion

A high-performance Analog Front-End (AFE) for EEG Alpha wave extraction was successfully designed, simulated, and validated. By utilizing a modular architecture of cascaded second-order Sallen-Key and Butterworth filters, the system achieves a balance between high gain, stability, and extreme noise rejection.

Summary of Achievements:

- **Precision Amplification:** Boosted microvolt-level neural signals to a measurable 65mV range.
- **Artifact Neutralization:** Successfully rejected 300mV DC offsets and 500mV common-mode interference without stage saturation.
- **ADC Compatibility:** Integrated a precise 1.65V level-shifter to ensure compatibility with 3.3V microcontroller ADCs.

The simulation results confirm that the circuit exceeds all *Analogverse* test criteria. Future work will focus on PCB layout optimization to minimize trace-induced noise and the integration of a digital FFT algorithm for real-time relaxation state detection.

The results of our work are attached with the drive link below:

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